

Name: _____
(sign the honor code at the end)

Astronomy 101 Pre-lab Exercise #5: Surfaces of the Terrestrial Planets

1. Sketch an impact crater and label the main parts of the crater (rim, floor, ejecta, central peak). Google “crater features” on the internet and look at images to get info on this. You can sketch either a side or face-on view.

2. Below are images of two different regions on the moon. What can you say about the relative ages of these two surfaces? State your reasoning.



Region A



Region B

3. (a) How can craters be used to estimate the age of a planetary surface? (b) For which planets in this lab activity will crater counting work the best? Why?

4a. What is the density of a spherical object with a mass of 7.2×10^{24} kg and a radius of 7.4×10^6 m? (Eqn 1)

4b. What is the escape velocity for this object? (Eqn 3)

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Lab Partners/Table #: _____

Astronomy 101 Laboratory #5: Surfaces of the Terrestrial Planets

Introduction:

Early observations of Mercury, Venus, the Moon and Mars were made telescopically. Telescopic observations allowed astronomers to measure the orbital properties, sizes, and masses of these planets. Mercury is difficult to observe in part because it is only visible shortly before sunrise or after sunset. In addition, very little of Mercury's surface is visible in terrestrial telescopes. The surface of Venus is hidden by a layer of impenetrable clouds. Craters, mountains, plains, etc. are easily visible on the Moon. Large surface features are visible on Mars, including seasonal variations in the Martian ice caps. Radio telescopes are used to bounce radar off of the planets, measuring their distances and rotation periods.

Goals:

1. To develop an understanding of how planetary scientists study the planets.
2. To gain experience analyzing numerical data and constructing and testing hypotheses.

The write-up contains detailed information that you need to read before answering the questions. It is helpful to discuss the questions in your lab group (those at your table) as you proceed. There are three activities in this lab. Activity 1 is computational; activities 2 and 3 require some artistic effort. Use a pencil! Be as neat as possible. Try to do a good job and make a nice sketch that contains the important geologic details. Extra paper is available for sketching geologic maps and doing computations. Part of your grade will be based on the organization of your work. The instructor will hand out and review with you a sheet on cratering and one on geological processes.

Activity 1. Planetary Properties.

Any study of the planets begins by comparing physical properties, i.e., size, mass, density, distance, and surface gravity. Is there an atmosphere present? Does the planet have natural satellites? Does the planet have a magnetic field? In this activity we will compare several physical properties of the planets. Table 1 contains data needed for your computations.

If we examine the information in Table 1, we find that larger planets contain more mass. This, of course, is no surprise and tells us nothing that we didn't know before we started. It might be interesting to know about how tightly packed the planet's matter is. Density is a measure of how much mass is in a given volume of material. In other words, density (ρ) is equal to mass (m) divided by volume (V),

$$\text{Density} = \rho = m/V. \quad \text{For a spherical object, } V = 4/3 \pi r^3 \quad (1)$$

A. Density: Density of planetary materials: iron-nickel: $\rho=8000 \text{ kg/m}^3$, stone: $\rho=3000 \text{ kg/m}^3$,
ice: $\rho=1000 \text{ kg/m}^3$, gas: $\rho<1000 \text{ kg/m}^3$

Q1) Calculate the density for the planets listed in Table 1. Lab partners should split the work among themselves so that no one person has to do all of the calculations. However, two people in each group should do each calculation independently and then compare. Record results in the table. **Show an example of your work on the back of this page.**

Q2) Based on the average density that you calculated, what % of the planet is rock and what % is metal? (The instructor will show an example calculation. **Show your work on the back of this page.**)

Venus: % rock = _____ % metal = _____

Mars: % rock = _____ % metal = _____

Table 1. Planetary Data

	Mercury	Venus	Earth	Moon	Mars
Mass (kg)	3.303×10^{23}	4.869×10^{24}	5.976×10^{24}	7.349×10^{22}	6.421×10^{23}
Radius (m)	2.440×10^6	6.052×10^6	6.379×10^6	1.737×10^6	3.397×10^6
Density (kg/m ³)					
Distance from Sun (km)	57,910,000 (=0.387 AU)	108,200,000 (=0.723 AU)	149,600,000 (=1.000 AU)	149,600,000 (=1.000 AU)	227,940,000 (=1.524 AU)
Rotation Period (days)	58.6462	-243.0187	0.99727	27.32166	1.0260
Orbital period (days)	87.969	224.701	365.256	27.32166	686.98
Visual Geometric Albedo	0.10	0.65	0.37	0.12	0.15
Average Surface Temperature (K)	450	680	290	280	210
Atmospheric Composition	He - 42% Na - 42% O ₂ - 15% other 1%	CO ₂ - 96% N ₂ - 3+% traces of SO ₂ , H ₂ O, CO, Ar, NE, HCl, HF	N ₂ - 77% O ₂ - 21% Ar - 1% H ₂ O - variable CO ₂ - .03%	Xe traces	CO ₂ - 95.3% N ₂ - 2.7% Ar - 1.6% O ₂ - 1.6% H ₂ O - 0.03%
Pressure (atm)	1.0×10^{-15}	92	1	...	0.007
Magnetic field (Earth =1)	0.01	$< 1 \times 10^{-4}$	1	1×10^{-4}	0.002

B. Patterns in the Properties:

Q3) Examine the other information in Table 1, e.g., distance from Sun, surface temperature, etc. (a) Is there any systematic pattern (trend) that you detect for the planets (exclude the moon), or at least for 3 of the 4? Try to find two such trends, not including distance and orbital period (Kepler's 3rd law). (b) Construct a possible explanation for your observations.

C. Retaining an Atmosphere:

Surface gravity: A planet's radius (R_p) and mass (M_p) also affect its surface gravity and its ability to hold onto an atmosphere. Any massive object near the surface of a planet experiences a downward force due to gravity. Near the surface of the planet, this force is proportional to the planet's mass and inversely proportional to the square of the planet's radius. The acceleration due to gravity at a planet's surface is described by the mathematical expression,

$$g = GM_p/R_p^2, \text{ where } G = 6.673 \times 10^{-11} \text{ N m}^2/\text{kg}^2 \quad (\text{N} = \text{newton, a unit of force}) \quad (2)$$

If M_p in units of kg and R_p is in units of m, then the answer will have the desired units of m/s^2 .

Escape speed: Another result that can be calculated from applying Newton's laws of motion to planetary problems is the escape speed. The escape speed is the minimum speed an object needs to attain at the planet's surface to successfully escape the gravitational attraction of the planet. The escape speed is described by the relation,

$$v_{\text{esc}} = (2 G M_p / R_p)^{1/2}. \quad (3)$$

Q4) (a) Calculate the surface gravity and escape speed for Mars and Venus. The surface gravity for Earth is 9.8 m/s^2 and the escape velocity is $1.12 \times 10^4 \text{ m/s}$ ($= 11.2 \text{ km/s}$). (Show all of your work below.)

Planet	$g \text{ (m/s}^2\text{)}$	$v_{\text{esc}} \text{ (m/s)}$
Mars		
Venus		

(b) Why would one expect the surface gravity and the composition of a planet's atmosphere to be related?

(c) What other properties might affect the make up of a planet's atmosphere?

Activity 2. Impact craters.

In this activity we will examine several images of planetary surfaces. Log on to the computer, using your regular username and password. Click on: *Computer*, then *Shared drive S*. Then go to *Academic\PHYSICS\A101L\PLANSCI*. You will see listed folders for the various terrestrial. To view the impact crater images, double click on a file name. Each lab group has two computers at its disposal so that you can view two different images at a time. No computer should be idle! Crater images for all four terrestrial planets and the Earth's moon are present.

Geologic maps and identifying crater morphology.

In the *Venus* subdirectory is a pair of images *DANI* and *DANII*. The *DANI* image is an image of Danilova Crater on Venus. The crater is 49 km (30 miles) in diameter and has a prominent central peak, a crater wall, a crater floor, an ejecta blanket and crater outflow deposits. The image *DANII* is a geologic map that identifies these regions. Planetary scientists construct geologic maps of a planet's surface as the first step in studying a planet's geology. It's a straightforward process. One draws an outline of each distinct feature of the surface and identifies what it might be. Later information from images at other wavelengths is combined with these to build up a detailed description of the surface. The Danilova Crater image and map are your introduction to constructing geologic maps of planetary surfaces. (A similar image map pair exists for the *YABL* and *YABLL* images of Yablochkina Crater (63 km or 39 mile diameter).)

Q5) Construct (sketch) a geologic map for **three craters**. Each crater should be from a **different planet**.

Go back up one directory and pick one from each planet. I suggest that you use the list below.

Describe what you observe about crater shape, an ejecta blanket, etc. for each crater. Use the following coding for the different parts: CP = central peak, CF = crater floor, CW = crater wall, CE = crater ejecta, CO = crater outflow.

Use at least a half-sheet of paper per drawing. (All need not be present in every crater.)

<u>Mercury:</u>	<u>Venus:</u>	<u>Earth:</u>	<u>Moon:</u>	<u>Mars:</u>
Mercury1 (d=40 km)	Addams (90 km)	Clearwat (10 km)	arch	Outflow
Mercury5	alcott (63 km)	Meteor	copernic (93 km)	Network
Caloris2	cleo (105 km)	Wolf (1 km)	aislands	
Fault	Golubnew	gosses (22 km)	plaskett	
	Mead		AS15_M3	
			Farside	
			LOIII113	

Be sure to explicitly list the name of the crater and the planet on which it is found.

Activity 3. Which came first? Relative age and Superposition

When you're exploring the planets, it's a good idea to understand how a landscape developed and what processes are involved in shaping the surface of a planet. **Superposition** is a simple way to figure out a sequence of events. Most of the time younger things are on top of older things, and younger or more recent events affect the appearance of older features.

Q6) Examine the image *VALLES_M* in the *MARS* subdirectory. These images are of the Mars surface. They contain a variety of geological features: craters, landslides, flow channels, scarps, ejecta blankets, lava flows, slump blocks, and debris aprons. (Not all features are in each image.) Refer to the diagrams provided by the instructor for schematic representations of these geologic features.

(a) Make a geologic map of the image.

(b) Label the major features (use the list above) **and (c) then list them in the order that you think they were formed based on the idea of superposition. Give reasons for your order.**

Q7) What are some of the things that you have learned about planetary surfaces from carrying out this lab?

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